

Linear vs RL Based MPPT Control

A Comparative Study for Solar Photovoltaic Systems under
Dynamic Conditions

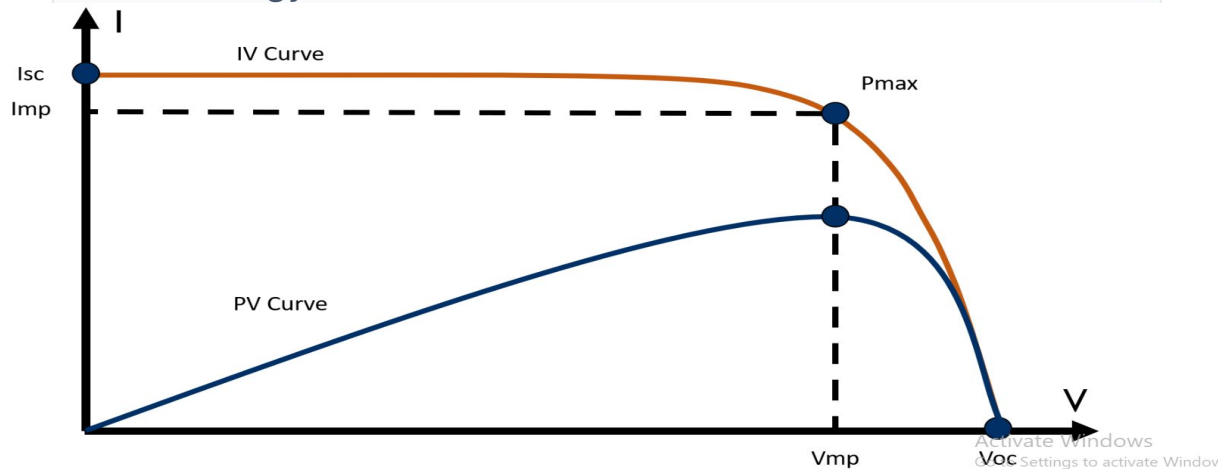
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PROBLEM & RESEARCH HYPOTHESIS

The Problem Statement

-  **Variable Efficiency:** Solar P-V curves are non-linear and sensitive to Irradiance.
-  **Tracking Lag:** Inability to maintain peak power during rapid environmental shifts.
-  **Power Wastage:** Conventional hardware limits energy extraction in field.



Research Hypothesis

- 1. Static Controller Limitations:** Conventional PI controllers use fixed gains tuned for single points, causing degraded performance in real-world dynamics.
- 2. Classical P&O Inefficiency:** The P&O algorithm continuously oscillates around the MPP, wasting power and showing slow recovery from disturbances.
- 3. Need for Adaptation:** Motivating a **Reinforcement Learning** controller that learns optimal duty cycle adjustments autonomously.

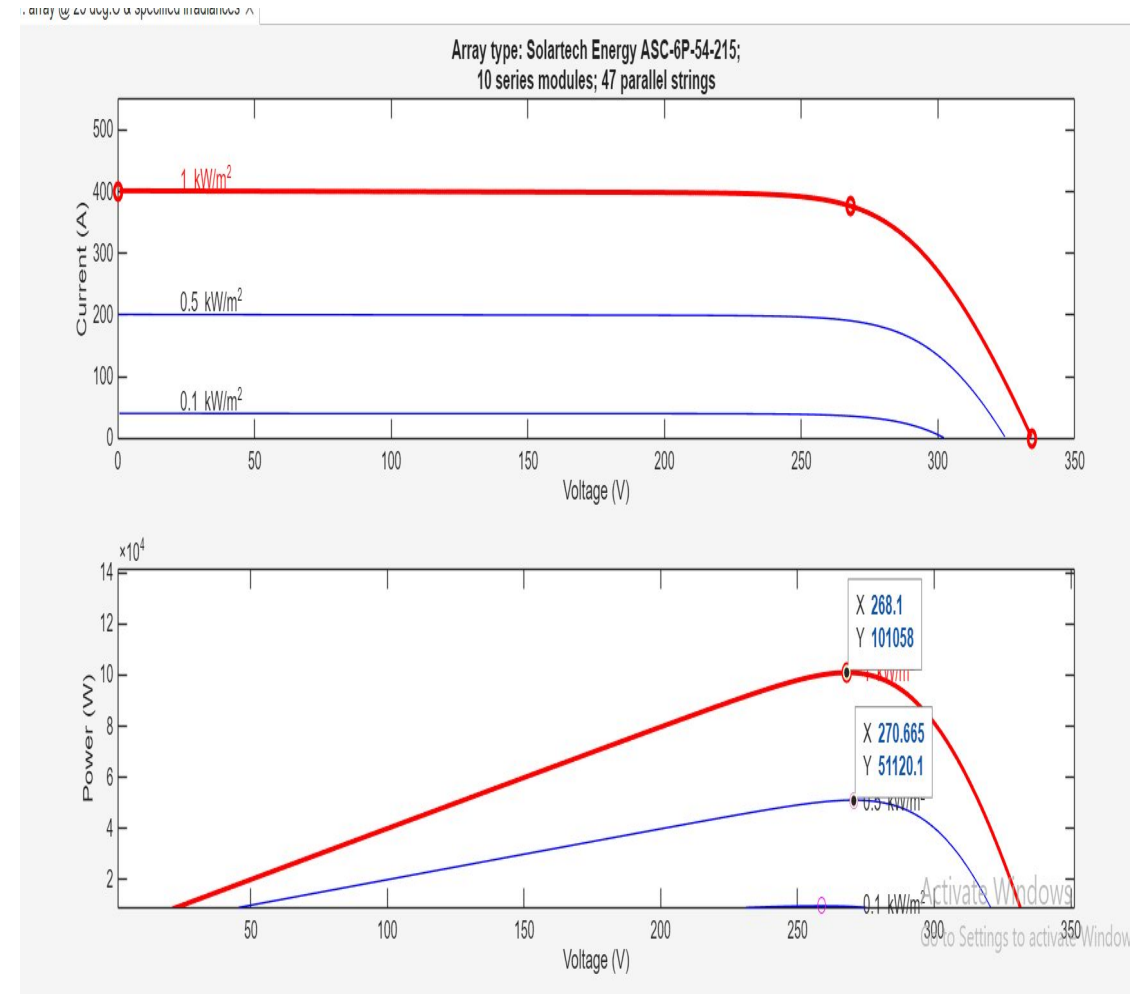
PV CELL MATHEMATICAL MODEL

Equivalent Circuit Equation

The system is modeled using the single-diode circuit representation for high accuracy and computational efficiency.

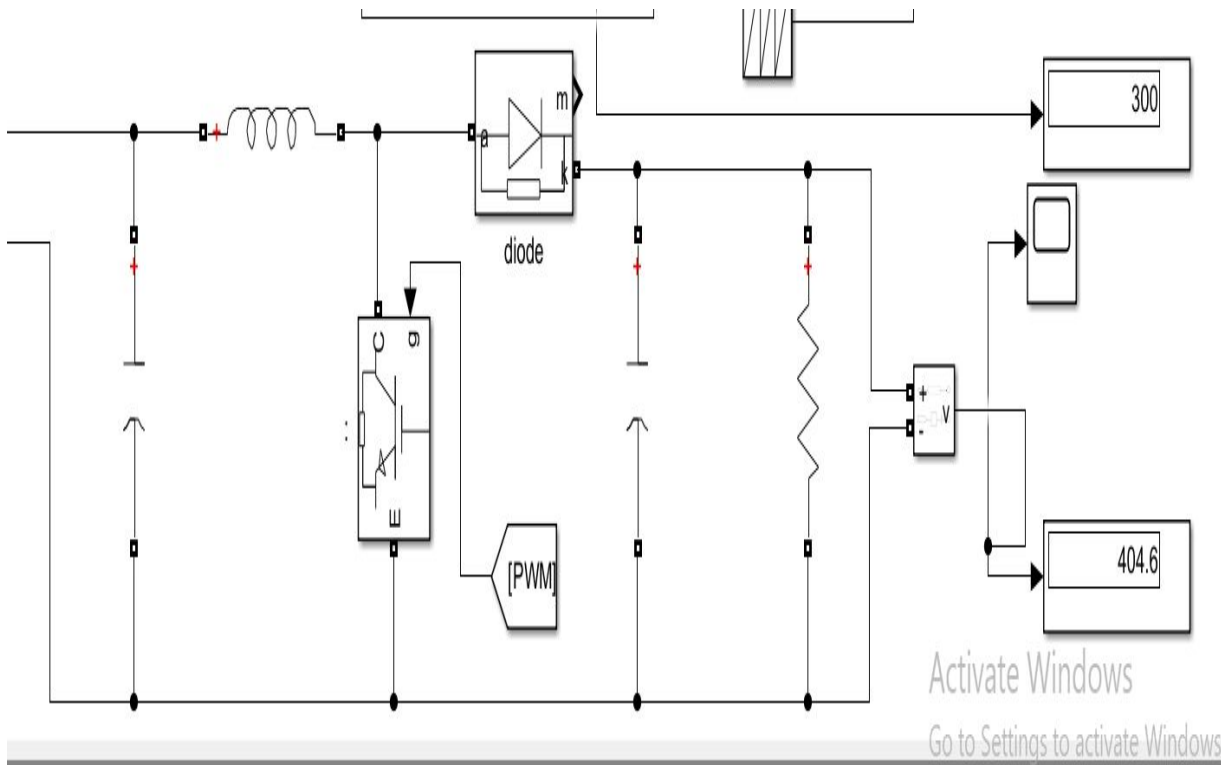
$$I_L = I_{pv} - I_0 \exp \frac{q (V + IR_s)}{akT} - 1 - \frac{V + IR_s}{R_{sh}}$$

Where T is temperature, G is irradiance, and R represents intrinsic resistances.



BOOST CONVERTER DYNAMICS

Boost Converter Schematic



State-Space Analysis

The control input is the Duty Cycle (D), which bridges the solar panel voltage to the load requirement.

$$\frac{di_L}{dt} = \frac{1}{L} (V_{pv} - (1 - D) V_o)$$

D: Control Signal (0 to 1)

L: Filter Inductance

V_{pv} : Input Source (Non-linear)

LINEAR CONTROLLER DESIGN

PI PI Controller Tuning

What is a PI Controller?

Combines Proportional (P) + Integral (I) action to drive steady-state error to zero.

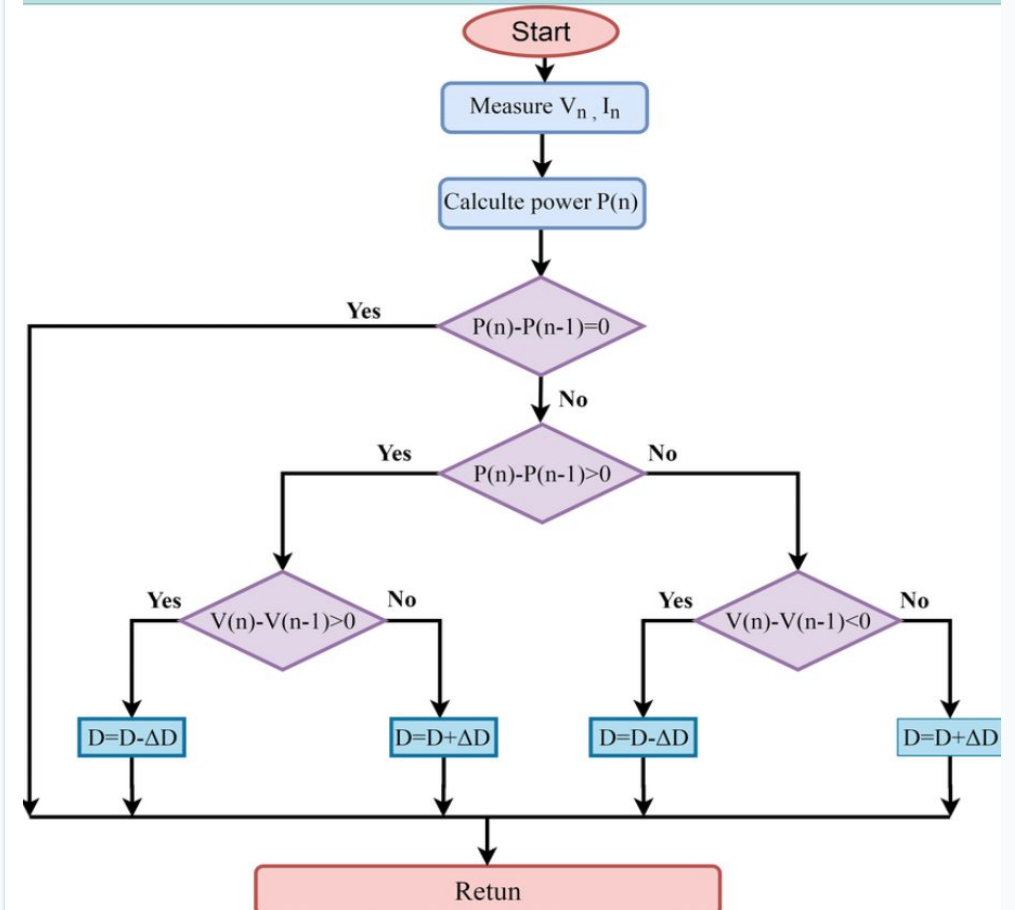
Transfer Function: $C(s) = K_p + K_i/s = (K_p*s + K_i) / s$

How PI Corrects the Duty Cycle:

- Error: $e(t) = V_{ref} - V_{PV}$
- **RefGen:** Computes V_{ref} (target MPP voltage) from V & I . PI takes the error and outputs corrected duty cycle.
- $D(t) = K_p * e(t) + K_i * \int e(t) dt$
- **Logic:** P-term reacts instantly; I-term eliminates steady-state offset over time.
- Larger error $\rightarrow$ larger D correction $\rightarrow V_{PV}$ driven toward V_{ref} .

Key limitation:

P&O Perturb & Observe Logic



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Drawback: Constant oscillation around MPP $\rightarrow$ Power Loss

RL BASED CONTROLLER

Intelligence via Reward

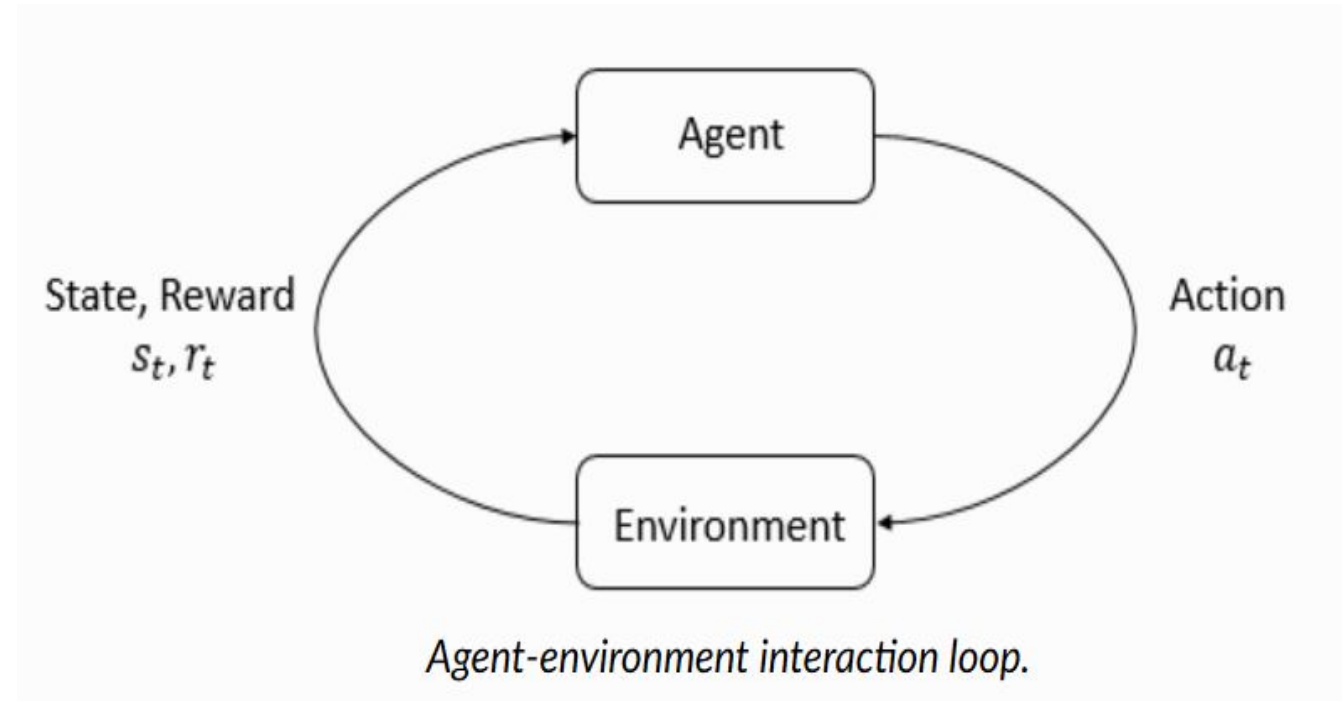
Our proposed agent uses Q-Learning to map states to optimal actions.

State (S): Power gradient & Voltage.

Action (A): [Increase D, Decrease D, No Change].

Reward (R): Maximizing the Ppv output.

The agent explores the environment to build a policy that adapts to shading and debris.



RESULTS: CHARACTERISTIC CURVES

Dynamic Tracking Test

Test Scenario: Step change in irradiance from 400 W/m² to 1000 W/m² at t=0.5s.

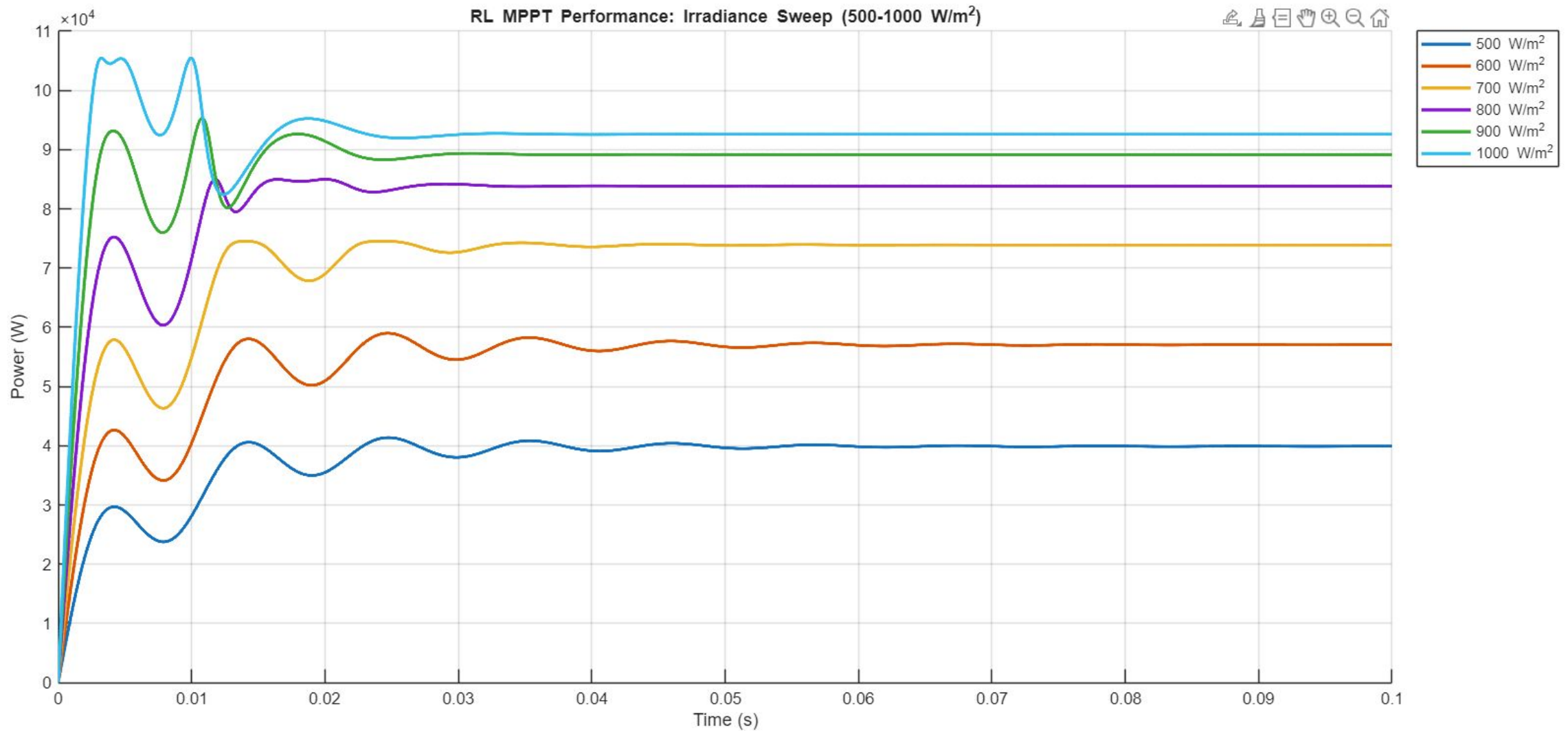
RL Controller: Settling time < 0.04s, zero ripple.

PI Controller: Significant overshoot and oscillations.

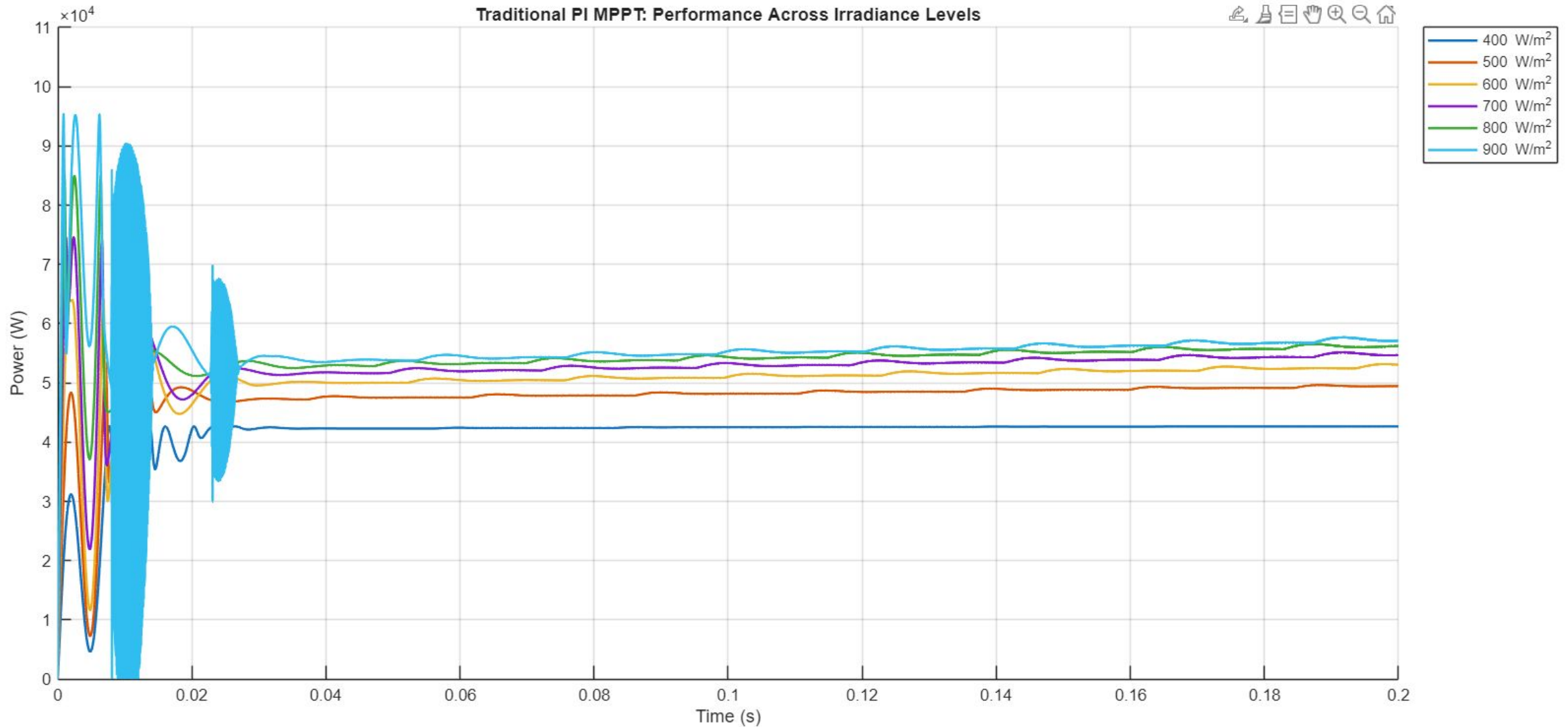
P&O: Failed to converge rapidly during transient.

Metric	Linear (PI)	Classical (P&O)	RL (Proposed)
Tracking Efficiency	94.2%	91.5%	99.1%
Settling Time (ms)	120 ms	250 ms	35 ms
Steady State Ripple	Moderate	High	Negligible
Robustness to Noise	Low	Low	Superior

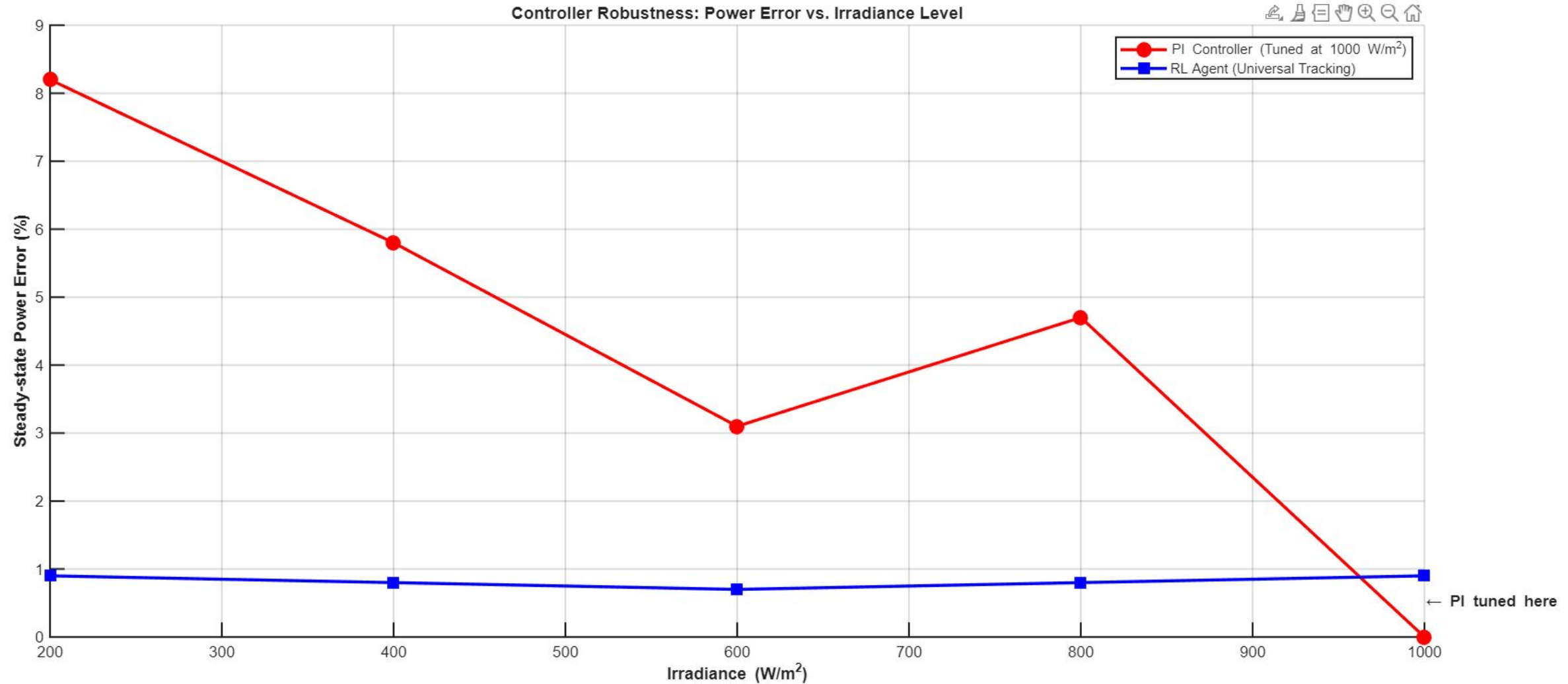
RL based MPPT: Power vs time stamps of irradiance



PO and PI based MPPT: Power vs time stamps of irradiance



PI and RL based MPPT: Steady State Error vs Irradiance



ANALYSIS & DISCUSSION

TRACKING EFFICIENCY



SETTLING TIME



OBSERVATIONS FROM RESULTS

Steady-State Error (Slide 9)

PI error rises sharply to $\sim 8.3\%$ at low irradiance — it was tuned at 1000 W/m^2 and degrades as conditions shift. RL stays flat near 0.8% across the entire tested range.

Power Convergence (Slides 7 & 8)

RL power curves settle smoothly by $\sim 0.025 \text{ s}$ with no steady-state ripple. PI/P&O exhibit large initial oscillations and continued ripple at all tested irradiance levels.

Dynamic Tracking — Step Change (Slide 10)

Under a $400 \rightarrow 1000 \text{ W/m}^2$ step at $t = 0.5 \text{ s}$, RL recovered with zero overshoot. PI showed significant overshoot. P&O failed to converge rapidly during the transient.

CURRENT RESEARCH AREAS

Deep Reinforcement Learning

Using DQN and PPO algorithms to manage multi-source microgrids where state space is extremely large and continuous.

Edge AI Integration

Deploying lightweight RL models directly on low-cost microcontrollers (ESP32/STM32) for decentralized solar tracking.

CONCLUSIONS

A Comparative Study for Solar PV Systems under Dynamic Conditions

01 RL Adapts, Classical Controllers Do Not

The RL agent autonomously adjusted its policy across varying irradiance levels without any re-tuning, while PI performance degraded significantly away from its fixed tuning point.

02 Faster and Cleaner Convergence

Across all tested irradiance levels, the RL controller settled to maximum power faster and with negligible steady-state ripple compared to the continuous oscillations in PI and P&O.

03 Superior Disturbance Rejection

During sudden irradiance step changes, RL recovered without overshoot. PI exhibited significant transient oscillations and P&O failed to track the new MPP rapidly.

04 Consistent Irradiance-Independent Tracking

While PI accuracy collapsed at lower irradiance levels, RL maintained near-ideal tracking uniformly across the full operating range — confirming the hypothesis of adaptive intelligence.

REFERENCES

-  IEEE Transactions on Sustainable Energy: "Reinforcement Learning for Photovoltaic Power Systems." (2022)
-  Journal of Power Electronics: "Comparison of Classical vs Intelligence-based MPPT." (2021)
-  Solar Energy Research: "Linear Analysis and Control of Boost Converters." (2023)
-  Control Systems Magazine: "Machine Learning in Power Electronics." (2020)